Success against cost: an operational efficiency of control

DRAFT, for discussion — Simulations 1 and 2, `task_003`

LLM-free simulator; analysis of 13 October 2026

Summary. Earlier information ratios (pair efficiency η, its variants, and the action-based η_U) were either flat by construction or close to zero. Following the operational proposal of the handoff document, we measure instead how much a controller changes the probability that the swarm's *collective decision* is its target (and the correct answer), and how many facts it posts to do so. Three results stand out. (i) **When no proof exists, success saturates with very little control**: the cheapest controller reaches the full gain, and most configurations spend two to four times more for nothing. (ii) **Speed is worth paying for only early**: the efficiency depends on when success is scored. (iii) **Overturning a proof has increasing returns**: concentrated control (random mixtures of strong control and none) beats every tested configuration.

1 The simulators in brief

24 agents, each given one true fact about `task_003` (three allocations; A0 is correct), reason as exact Bayesians and post facts on a shared board; a controller with a target (A0, truth; A2, false) posts facts from its own pool. **Simulation 1**: days and nights; each agent reads q posts a day, votes, and posts a fact supporting its vote; each night the controller reads q_c posts and, unless 75% of them already back its target (vote gate) or its target is proved (proof stop), posts b facts; the board is cleared nightly; memories fade at dawn (survival ρ). **Simulation 2**: no rounds; agents act at random times (rate 1), the controller at rate λ_c (one fact per message, at most 30, until $t = 30$; gate on or off); memories fade continuously; readers see the 48 most recent posts. Agents vote by probability matching; $q = 6$ throughout; 1,000 paired episodes per arm (silent, truth, false).

2 Measures

- **Collective decision** Y_T: plurality of the agents' last votes at time T; a tie between k answers counts $1/k$. **Target attainment**: Y_T = assigned target; **task correctness**: Y_T = A0.
- **Benefit** $\Delta_\pi = p_\pi - p_0$: change in the probability of success against the silent arm of the same episodes (scored with the same target).
- **Cost** C_π: facts actually posted up to T, mean per run (not the allowed budget).
- **Frontier** among the tested controller settings of one group (simulation, setup, ρ, target): $C^*(\delta) = \min\{C_\pi : \Delta_\pi \geq \delta\}$ and $\Delta^*(B) = \max\{\Delta_\pi : C_\pi \leq B\}$; with **mixtures** (one configuration drawn per run) the frontier is the concave hull through $(0, 0)$.
- **Efficiencies**: $\eta_{\text{cost}} = C^*(\Delta_\pi)/C_\pi$ (share of the spending that was necessary), $\eta_{\text{perf}} = \Delta_\pi/\Delta^*(C_\pi)$, and **gain per fact** Δ_π/C_π. Not defined where $\Delta_\pi \leq 0$ or its 95% interval includes 0.

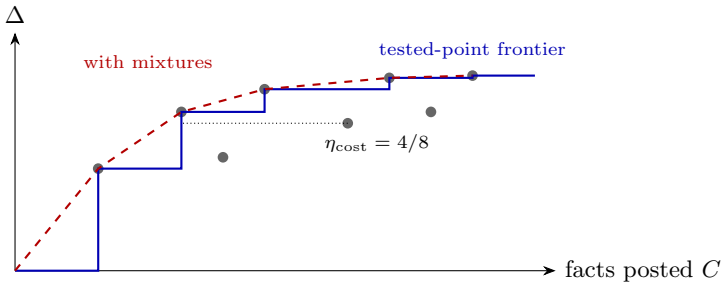


Diagram 1. Each point is a configuration. The configuration at cost 8 reaches $\Delta = 0.65$, which a configuration costing 4 already reaches: $\eta_{\text{cost}} = 0.5$.

3 Success against facts posted

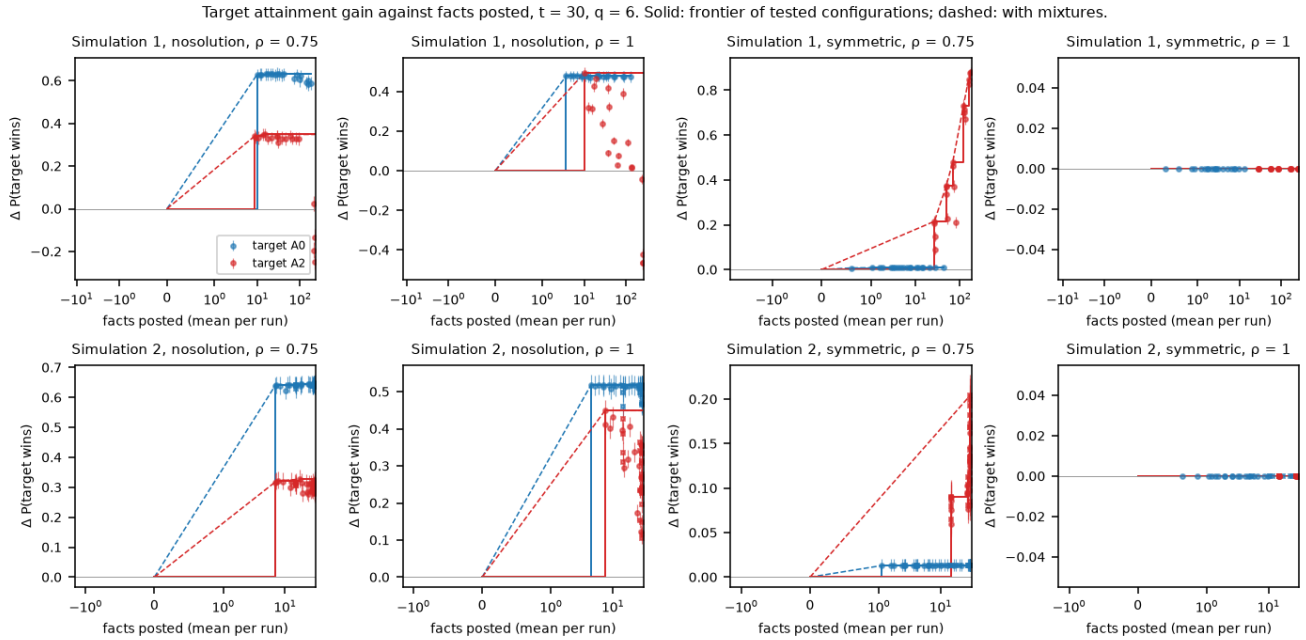


Figure 1: Change in the probability that the controller's target wins, $t = 30$, against facts posted (log scale), every configuration with its 95% interval (blue: target A0; red: target A2; Simulation 2: circles gate on, squares gate off). Solid: tested-point frontier; dashed: with mixtures.

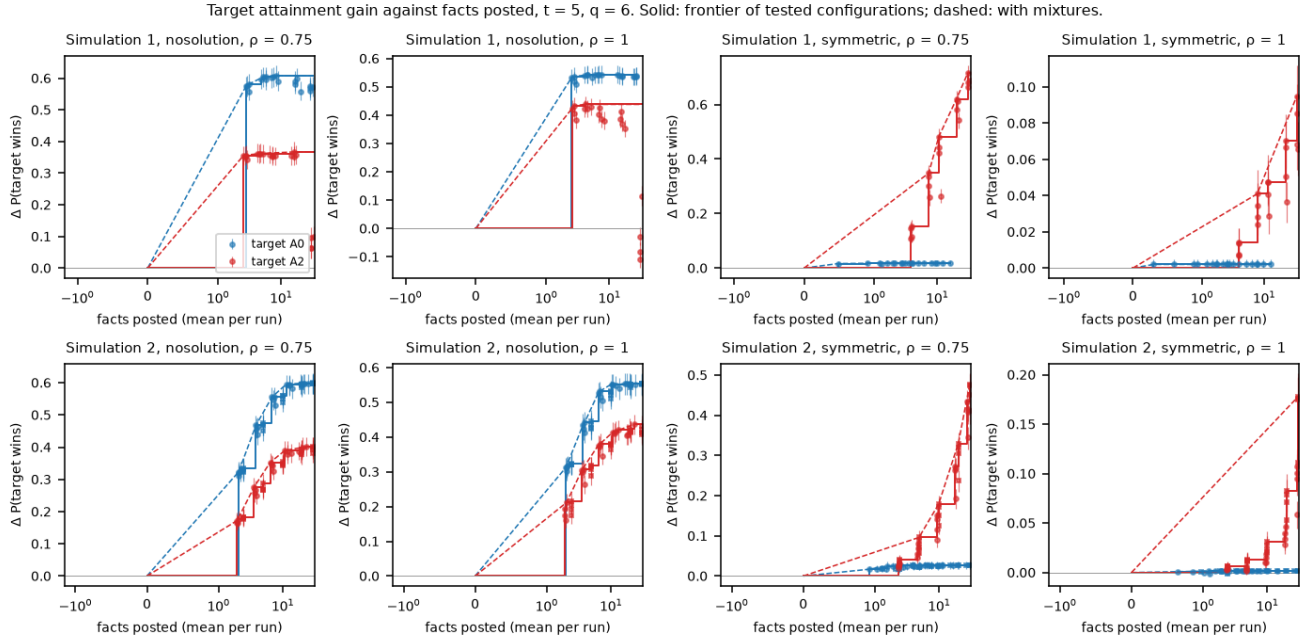


Figure 2: As Figure 1, at $t = 5$.

Table 1: Overview at $t = 30$ (target attainment; groups with a measurable effect). p_0: silent probability that the target wins; frontier: configurations with $\eta_{\text{cost}} = 1$.

	setup	ρ	target	p_0	range of Δ	facts	on frontier	median η_{cost}	min
Sim. 1	nosolution	0.75	A2	0.63	-0.25 to +0.35	9–221	3/20	0.30	0.09
Sim. 1	nosolution	0.75	A0	0.36	+0.59 to +0.63	10–176	6/20	0.43	0.06
Sim. 1	nosolution	1	A2	0.48	-0.47 to +0.49	10–248	1/20	0.26	0.07
Sim. 1	nosolution	1	A0	0.52	+0.47 to +0.48	4–128	1/20	0.18	0.03
Sim. 1	symmetric	0.75	A2	0.01	+0.09 to +0.88	27–173	9/20	0.99	0.34
Sim. 2	nosolution	0.75	A2	0.64	+0.28 to +0.33	7–30	4/40	0.26	0.23
Sim. 2	nosolution	0.75	A0	0.35	+0.62 to +0.65	7–30	8/40	0.48	0.23
Sim. 2	nosolution	1	A2	0.52	+0.11 to +0.45	8–30	1/40	0.27	0.25
Sim. 2	nosolution	1	A0	0.48	+0.46 to +0.52	4–30	1/40	0.17	0.15
Sim. 2	symmetric	0.75	A2	0.01	+0.06 to +0.20	15–30	15/40	0.99	0.49

Table 2: Least facts needed to raise the target's chance of winning by δ ($t = 30$): tested configurations / with mixtures; – : not reached.

	setup	ρ	target	$\delta = 0.05$	0.1	0.2	0.3
Sim. 1	nosolution	0.75	A2	8.7 / 1.3	8.7 / 2.5	8.7 / 5.1	8.7 / 7.6
Sim. 1	nosolution	0.75	A0	9.8 / 0.8	9.8 / 1.6	9.8 / 3.1	9.8 / 4.7
Sim. 1	nosolution	1	A2	10.1 / 1.0	10.1 / 2.1	10.1 / 4.1	10.1 / 6.2
Sim. 1	nosolution	1	A0	3.7 / 0.4	3.7 / 0.8	3.7 / 1.6	3.7 / 2.4
Sim. 1	symmetric	0.75	A2	26.9 / 6.2	26.9 / 12.5	26.9 / 25.0	49.3 / 38.9
Sim. 2	nosolution	0.75	A2	6.9 / 1.1	6.9 / 2.2	6.9 / 4.3	6.9 / 6.5
Sim. 2	nosolution	0.75	A0	6.9 / 0.5	6.9 / 1.1	6.9 / 2.2	6.9 / 3.3
Sim. 2	nosolution	1	A2	7.5 / 0.8	7.5 / 1.7	7.5 / 3.3	7.5 / 5.0
Sim. 2	nosolution	1	A0	4.4 / 0.4	4.4 / 0.9	4.4 / 1.7	4.4 / 2.6
Sim. 2	symmetric	0.75	A2	14.6 / 7.4	27.0 / 14.7	30.0 / 29.4	– / –

4 Phase diagrams

Each panel: tiles over the controller's reading budget q_c (rows) and its rate λ_c (Simulation 2) or facts per night b (Simulation 1); columns: Δ, facts posted, gain per fact and η_{cost}, for each target; dash: not defined.

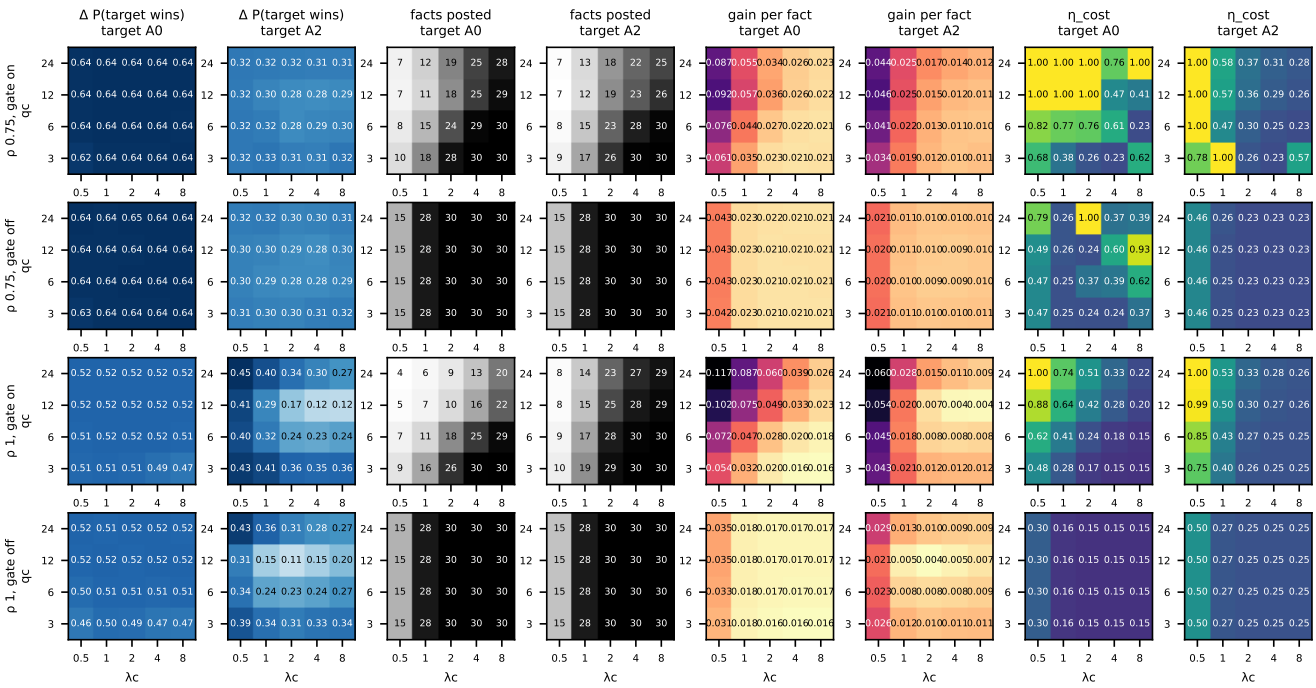


Figure 3: Simulation 2, nosolution, $t = 30$.

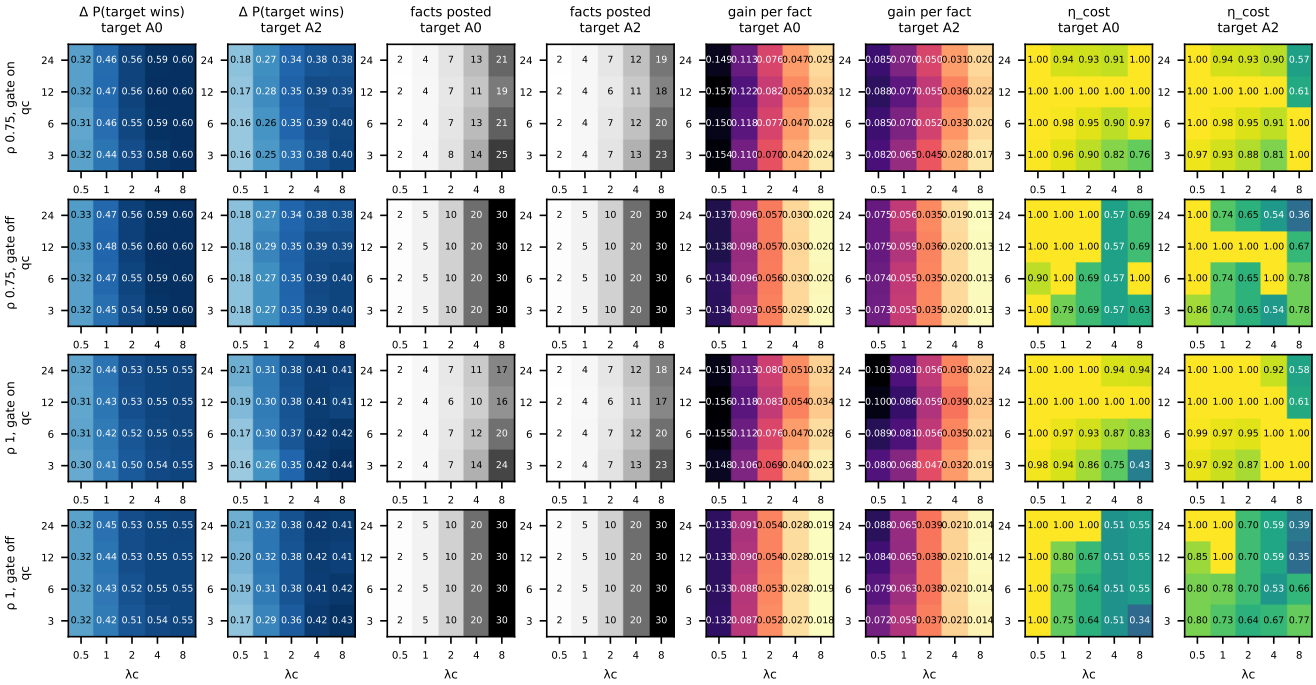


Figure 4: Simulation 2, nosolution, $t = 5$.

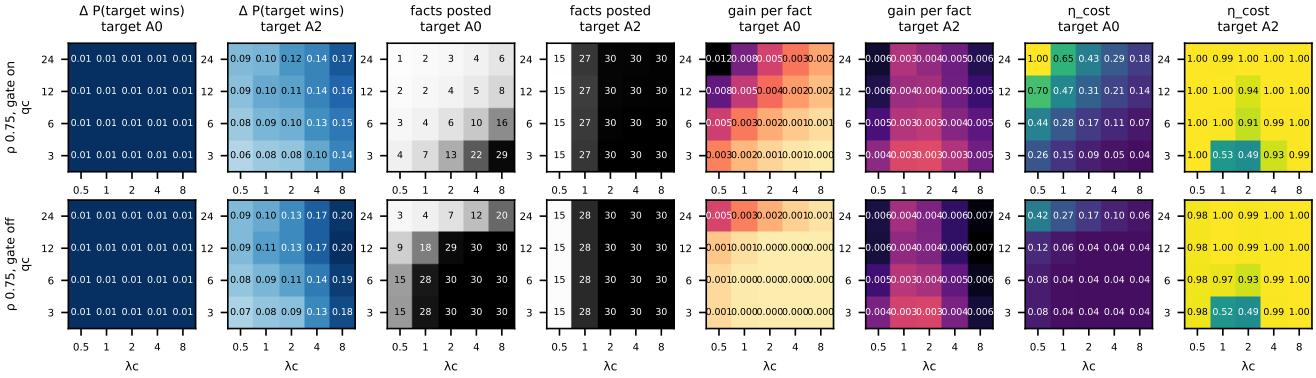


Figure 5: Simulation 2, symmetric, $\rho = 0.75$, $t = 30$ (with full memory every agent proves A0 and nothing changes).

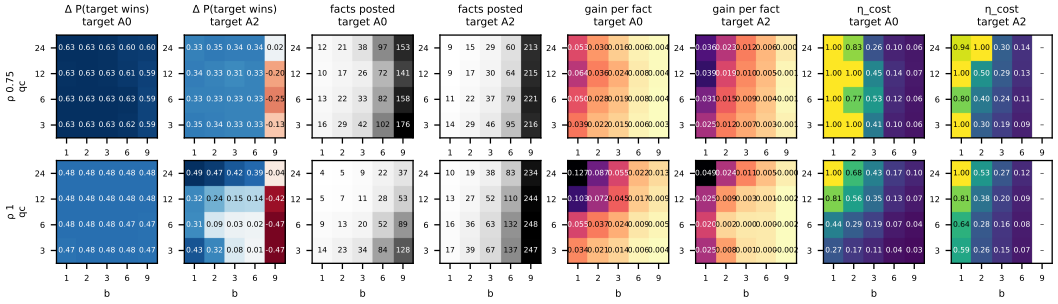


Figure 6: Simulation 1, nosolution, day 30.

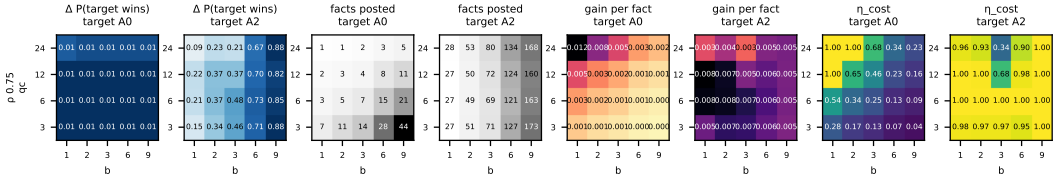


Figure 7: Simulation 1, symmetric, $\rho = 0.75$, day 30.

5 Manipulation against correctness

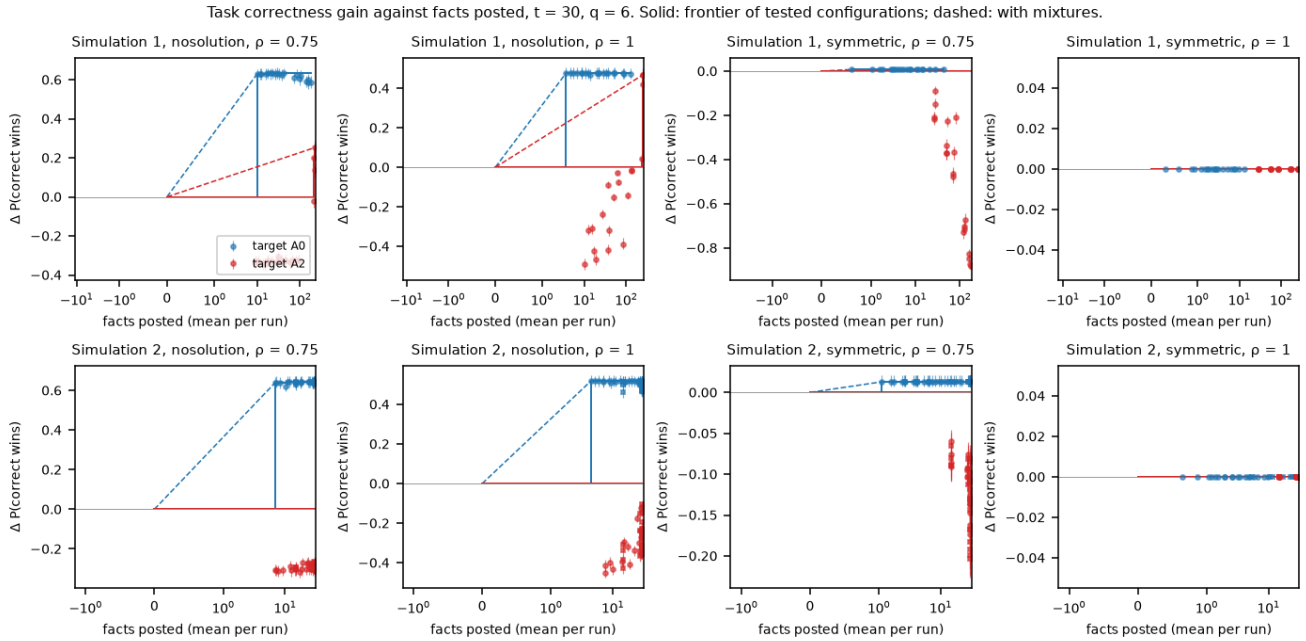


Figure 8: As Figure 1, for task correctness (A0 wins). For the false controller the change is negative: what it gains in target attainment is lost in correctness.

6 Findings

1. **Nosolution: success saturates with very little control.** At $t = 30$ every configuration reaches about the same gain (Simulation 2, $\rho = 0.75$: +0.28 to +0.33 for target A2, +0.62 to +0.65 for A0) whatever it spent (7–30 facts). The cheapest tested controller — slowest rate, gate on — reaches every threshold up to +0.3; most others are dominated (median η_{cost} 0.26 and 0.48). In Simulation 1 large batches waste even more (η_{cost} 0.03–0.09 at $b = 6$–9). The waste is posting after the decision is settled.
2. **Speed pays only early.** At $t = 5$ faster controllers win more (target A2: +0.17 with 2 facts, +0.39 with 18–30; Figure 2, 4) with diminishing returns; by $t = 10$ slow controllers catch up.
3. **Overturning a proof has increasing returns** (symmetric, target A2): gains grow faster than facts (Simulation 1 up to +0.88 at 173 facts). The frontier is convex, so mixtures of no control and strong control beat every tested configuration (to reach +0.1: 27 facts tested, 15 with mixtures, in Simulation 2). Most tested points lie on the frontier ($\eta_{\text{cost}} \approx 1$): the scalar saturates here, the frontier's shape does not.
4. **Manipulation costs correctness one for one**: for the false controller the fall in task correctness equals the gain in target attainment (median ratio 0.96–1.00; Figure 8).
5. **Ceilings and harm**: symmetric with target A0, silence already wins 99–100% of runs; Simulation 1, nosolution, large b: negative gains (false controller down to −0.47), no efficiency.

Table 3: Precision: root-mean-square error of Δ and of η_{cost} (median over configurations, $t = 30$) from 50, 100 or 300 paths per arm (200 subsets each), against 1,000 paths.

	outcome	Δ, 50	η, 50	Δ, 100	η, 100	Δ, 300	η, 300
Sim. 1	target attainment	0.057	0.063	0.039	0.035	0.021	0.017
Sim. 1	task correctness	0.059	0.038	0.041	0.028	0.021	0.017
Sim. 2	target attainment	0.025	0.033	0.017	0.021	0.009	0.008
Sim. 2	task correctness	0.022	0.017	0.016	0.012	0.008	0.006

7 Caveats and next steps

- The frontier is relative to the tested settings and **not identified below the cheapest tested cost** (about 7 facts in Simulation 2): the dashed hull assumes mixing with no control. Sample smaller budgets (1, 2, 3,

5 messages) and slower rates.

- For manipulation against a proof, a **burst controller** (spending its budget in concentrated episodes) is the natural next policy, given the increasing returns.
- Cost counts facts posted; sensing (posts read) is recorded but not yet included.
- Earlier attempts, for the record: the pair efficiency η (and η_{JS}, η_3) measures the symmetry of the two controllers, not size; the action-based η_U is ≈ 0 everywhere (control is cumulative, not punctual) and not estimable at 100 paths.

Where the results are. Code and notes: `analysis/task003_llm_free/efficiency_estimates/` (`frontier.py`, `2026-10-13_note_frontier.md`); tables: `results/efficiency_estimates/2026-10-13/` (`frontier_points.csv`, `frontier_thresholds.csv`, `frontier_downsample.csv`).